Topic 11 Cavity Method for Wigner Matrices

Recall from last lecture:

J is a random symmetric matrix with $J_{ij} \sim \begin{cases} \mathcal{N}(0, \sigma^2/N) & i \neq j \\ \mathcal{N}(0, 2\sigma^2/N) & i = j \end{cases}$

$$\rho_J(z) \equiv \frac{1}{N} \sum_{i=1}^{N} \delta[\lambda - \lambda_i] \qquad \lambda_i = \text{eigenvalues of } J$$

eigenvalue density $\rho(\lambda)$ of J

$$\mathcal{E}_J(z) \equiv \frac{1}{N} \mathrm{Tr}\left[zI - J\right]^{-1}$$

Stieltjes transform

or electric field at location $z \in \mathbb{C}$ of a set of point charges located at λ_i

- at large $N \to \infty$ $\mathcal{E}_J(z)$ concentrates about its mean $\mathcal{E}(z) = \langle \mathcal{E}_J(z) \rangle_J$

- we showed via replica theory that $\mathcal{E}(z)$ satisfies a self consistent equation

$$\boxed{\mathcal{E}(z) = \frac{1}{z - \sigma^2 \mathcal{E}(z)}}$$

today we will derive this equation via a cavity method without replicas...

First an aside: Stieltjes transform is a moment generating function:

$$\mathcal{E}_J(z) = \frac{1}{N} \mathrm{Tr} \frac{1}{z - J}$$

$$= \frac{1}{z} \frac{1}{N} \mathrm{Tr} \frac{1}{1 - J/z}$$

$$= \frac{1}{z} \sum_{k=0}^{\infty} \frac{1}{z^k} \frac{1}{N} \mathrm{Tr}\, J^k$$

$$\boxed{\begin{aligned} \mathcal{E}_J(z) &= \sum_{k=0}^{\infty} \frac{m_k}{z^{k+1}} \\ m_k = \frac{1}{N} \mathrm{Tr}\, J^k &= \int d\lambda\, \lambda^k \rho(\lambda) \end{aligned}}$$

- $\mathcal{E}(z)$ is a moment generating function for moments of $\rho(\lambda)$
- knowledge of $\mathcal{E}(z)$ near $z \to \infty$ is equivalent to knowledge of moments of charge density (like a multipole expansion in electromagnetism)

a simple moment based approach to get Wigner's semicircle law:

step 1: Compute $\langle m_k \rangle_J = \frac{1}{N} \langle \mathrm{Tr}\, J^k \rangle_J = \begin{cases} 0 & \text{for } k \text{ odd} \\ \underbrace{\frac{\sigma^{2k}}{k/2+1} \binom{k}{k/2}} & \text{for } k\text{-even} \end{cases}$

catalan numbers = # noncrossing pair pairings

$$\underbrace{J_{ij}\,J_{ke}}\;\underbrace{J_{mn}\,J_{no}}$$ (wick contractions)

or

$$\underbrace{\;\sqcup\;\sqcup\;\sqcup\;}$$

Step 2: realize that the wigner semicircular law for $\rho(x)$ is the only distribution that has these moments

now back to the cavity method using $\mathcal{E}_J(z)$

break the matrix into 1 component and N-1 components

let $M \equiv zI - J \qquad E \equiv M^{-1} \qquad \mathcal{E}_J(z) = \frac{1}{N}\,\mathrm{Tr}\,E$

$$M = \begin{array}{c|c} & 1 \;\; 2 \;\; 3 \cdots N \\ \begin{matrix}1\\2\\3\\\vdots\\N\end{matrix} & \left[\begin{matrix} \cdot & \text{---} \\ | & [\;\;] \end{matrix}\right] \end{array}$$

<u>Strategy:</u>

- find a relation between Stieltjes transform of a Wigner matrix J of size $N-1$ and N
- in the large N limit both transforms are the same
- this yields a self consistent equation for $\mathcal{E}(z)$

to find the relation between N and N-1 use schurs complement formula:

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \qquad E = M^{-1} = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}$$

decomposing of any matrix M and its inverse E into <u>blocks</u> of arbitrary size

then $$\boxed{E_{11}^{-1} = M_{11} - M_{12}\,M_{22}^{-1}\,M_{21}}$$

- apply this to our setting where block 1 is of size 1 and block 2 is of size N-1

$$\frac{1}{E_{11}(z)} = \underbrace{M_{11}(z)}_{\substack{z - J_{11} \\ \to z + \mathcal{O}(1/\sqrt{N})}} - \underbrace{M_{12}}_{J_{1i}}\,\underbrace{M_{22}^{-1}}_{E_{N-1}(z)}\,\underbrace{M_{21}}_{J_{j1}}$$

- take expected value of both sides over J:

① $$\left\langle \frac{1}{E_{11}(z)} \right\rangle_J \approx \frac{1}{\langle E_{11}(z)\rangle_J}$$

assume $E_{11}(z)$ converges to a deterministic value as $N\to\infty$ (ie fluctuations vanish as $N\to\infty$)

② $\langle M_{11}(z) \rangle_J = z$

③ $\sum_{i,j} \langle J_{1i} \, E_{N-1}(z)_{ij} \, J_{ji} \rangle_J$

J_{1i} independent of $E_{N-1}(z)_{ij}$

ie the N-1 cavity system does not know about coupling to new spin!

$$= \sum_{i,j} \langle J_{1i} J_{1j} \rangle \langle E_{N-1}(z)_{ij} \rangle_J$$

$$= \sum_{i,j} \frac{\sigma^2}{N} \delta_{ij} \langle E_{N-1}(z) \rangle_J$$

$$= \sigma^2 \sum_{i=1}^{N} \frac{1}{N} \langle E_{N-1}(z) \rangle_J$$

$$= \sigma^2 \frac{1}{N} \langle \mathrm{Tr}\, E_{N-1}(z) \rangle_J$$

$$= \sigma^2 \, \varepsilon(z) \; !!$$

So we have:

$$\frac{1}{\langle E_{11}(z) \rangle_J} = z - \sigma^2 \varepsilon(z)$$

now rotational inv of J

$\Rightarrow$ all diagonal elements $\langle E_{ii}(z) \rangle_J$ are the same

so $\langle E_{11}(z) \rangle_J = \frac{1}{N} \sum_{i=1}^{N} \langle E_{ii}(z) \rangle_J = \varepsilon(z)$!!

so we have

$$\frac{1}{\varepsilon(z)} = z - \sigma^2 \varepsilon(z)$$

or $$\boxed{\varepsilon(z) = \frac{1}{z - \sigma^2 \varepsilon(z)}}$$

self consistent equation for $\varepsilon(z)$ via cavity method!!